SHETH L.U.J. AND SIR M.V. COLLEGE

Implementation Of Research Using R

CODE:-

```
install.packages("caret")
install.packages("e1071")
install.packages("ggplot2")
install.packages("pROC")
install.packages("dplyr")
install.packages("smotefamily")


library(caret)
library(e1071)
library(ggplot2)
library(pROC)
library(dplyr)
library(smotefamily)


data <- read.csv("heart.csv")


data$target <- as.factor(data$target)


set.seed(123)

trainIndex <- createDataPartition(data$target, p = 0.7, list = FALSE)
train <- data[trainIndex, ]
test  <- data[-trainIndex, ]
```

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ROLL NO. S093
SUBJECT:- Data Analysis With SAS / SPSS / R

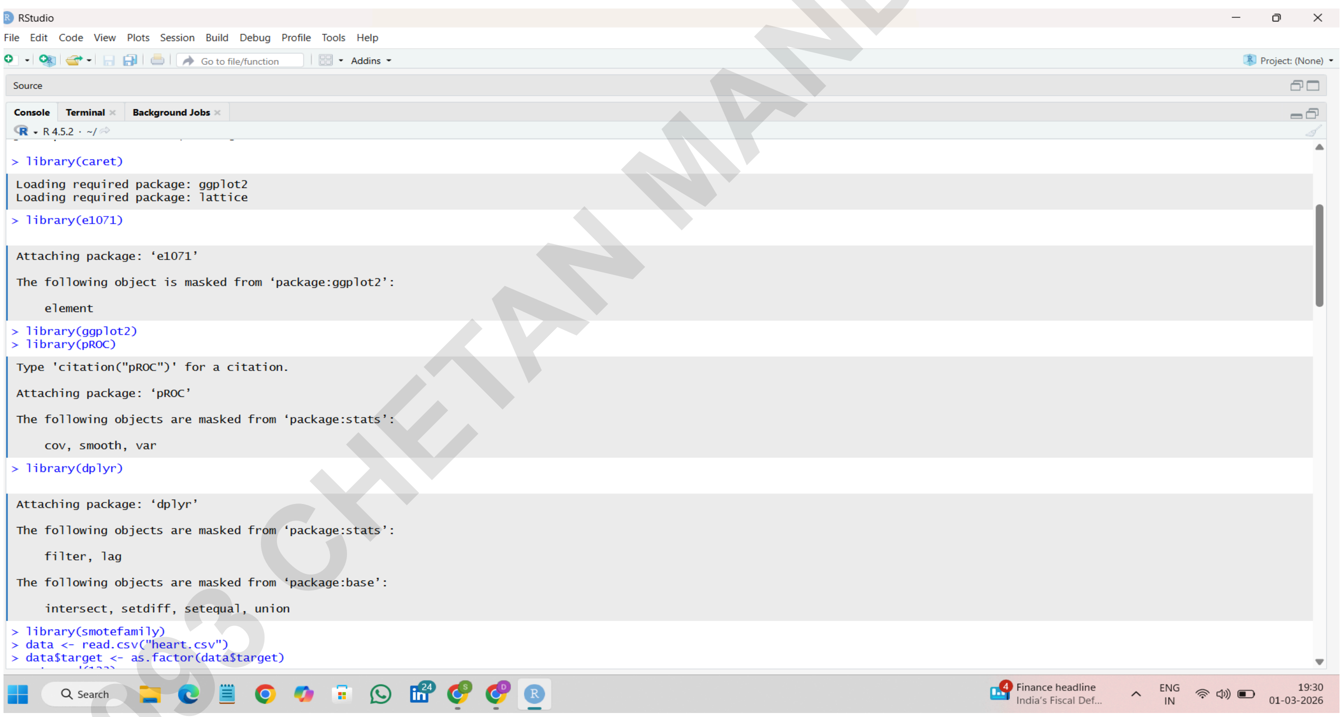
```
model_raw <- svm(target ~ ., data = train, probability = TRUE)

pred_raw <- predict(model_raw, test, probability = TRUE)

cm_raw <- confusionMatrix(pred_raw, test$target)

accuracy_raw <- as.numeric(cm_raw$overall["Accuracy"])

cat("\nAccuracy WITHOUT Preprocessing:", round(accuracy_raw,3), "\n")



preProc_impute <- preProcess(train, method = "medianImpute")
train_imputed <- predict(preProc_impute, train)
test_imputed  <- predict(preProc_impute, test)


preProc_scale <- preProcess(train_imputed, method = c("center", "scale"))
train_scaled <- predict(preProc_scale, train_imputed)
test_scaled  <- predict(preProc_scale, test_imputed)


train_x <- train_scaled[, colnames(train_scaled) != "target"]
train_y <- train_scaled$target


train_y_num <- as.numeric(as.character(train_y))


smote_output <- SMOTE(train_x, train_y_num, K = 5)

train_balanced <- smote_output$data
```

```
colnames(train_balanced)[ncol(train_balanced)] <- "target"


train_balanced$target <- as.factor(train_balanced$target)


model_processed <- svm(target ~ .,
                       data = train_balanced,
                       probability = TRUE)

pred_processed <- predict(model_processed,
                          test_scaled,
                          probability = TRUE)

cm_processed <- confusionMatrix(pred_processed, test_scaled$target)

accuracy_processed <- as.numeric(cm_processed$overall["Accuracy"])

cat("Accuracy WITH Preprocessing:", round(accuracy_processed,3), "\n")



accuracy_df <- data.frame(
  Model = c("Raw Data", "Preprocessed Data"),
  Accuracy = c(accuracy_raw, accuracy_processed)
)

ggplot(accuracy_df, aes(x = Model, y = Accuracy, fill = Model)) +
  geom_bar(stat = "identity", width = 0.6) +
  coord_cartesian(ylim = c(0,1)) +
  geom_text(aes(label = round(Accuracy,3)),
            vjust = -0.5, size = 5) +
```

```
  ggtitle("Impact of Data Preprocessing on Prediction Accuracy") +
  ylab("Accuracy") +
  xlab("Model") +
  theme_minimal() +
  theme(legend.position = "none")



prob_raw <- attr(pred_raw, "probabilities")[,2]

roc_raw <- roc(response = test$target,
               predictor = prob_raw,
               levels = c("0","1"),
               direction = "<")


prob_processed <- attr(pred_processed, "probabilities")[,2]

roc_processed <- roc(response = test_scaled$target,
                     predictor = prob_processed,
                     levels = c("0","1"),
                     direction = "<")


plot(roc_raw, col="red", main="ROC Curve Comparison")
plot(roc_processed, col="blue", add=TRUE)

legend("bottomright",
       legend=c("Raw Data","Preprocessed Data"),
       col=c("red","blue"),
       lwd=2)
```

```
cat("\nAUC Raw:", round(auc(roc_raw),3), "\n")
cat("AUC Processed:", round(auc(roc_processed),3), "\n")


cat("\nConfusion Matrix - Raw Data\n")
print(cm_raw)

cat("\nConfusion Matrix - Preprocessed Data\n")
print(cm_processed)
```

OUTPUT:-



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```
> library(smotefamily)
> data <- read.csv("heart.csv")
> data$target <- as.factor(data$target)
> set.seed(123)
> trainIndex <- createDataPartition(data$target, p = 0.7, list = FALSE)
> train <- data[trainIndex, ]
> test  <- data[-trainIndex, ]
> model_raw <- svm(target ~ ., data = train, probability = TRUE)
> pred_raw <- predict(model_raw, test, probability = TRUE)
> cm_raw <- confusionMatrix(pred_raw, test$target)
> accuracy_raw <- as.numeric(cm_raw$overall["Accuracy"])
> cat("\nAccuracy WITHOUT Preprocessing:", round(accuracy_raw,3), "\n")

Accuracy WITHOUT Preprocessing: 0.822
> preProc_impute <- preProcess(train, method = "medianImpute")
> train_imputed <- predict(preProc_impute, train)
> test_imputed  <- predict(preProc_impute, test)
> preProc_scale <- preProcess(train_imputed, method = c("center", "scale"))
> train_scaled <- predict(preProc_scale, train_imputed)
> test_scaled  <- predict(preProc_scale, test_imputed)
> train_x <- train_scaled[, colnames(train_scaled) != "target"]
> train_y <- train_scaled$target
> train_y_num <- as.numeric(as.character(train_y))
> smote_output <- SMOTE(train_x, train_y_num, K = 5)
There were 50 or more warnings (use warnings() to see the first 50)

> train_balanced <- smote_output$data
> colnames(train_balanced)[ncol(train_balanced)] <- "target"
> train_balanced$target <- as.factor(train_balanced$target)
> model_processed <- svm(target ~ .,
+                        data = train_balanced,
+                        probability = TRUE)
> pred_processed <- predict(model_processed,
+                           test_scaled,
+                           probability = TRUE)
> cm_processed <- confusionMatrix(pred_processed, test_scaled$target)
> accuracy_processed <- as.numeric(cm_processed$overall["Accuracy"])
> cat("Accuracy WITH Preprocessing:", round(accuracy_processed,3), "\n")
```

```
Accuracy WITH Preprocessing: 0.822
> accuracy_df <- data.frame(
+   Model = c("Raw Data", "Preprocessed Data"),
+   Accuracy = c(accuracy_raw, accuracy_processed)
+ )
> ggplot(accuracy_df, aes(x = Model, y = Accuracy, fill = Model)) +
+   geom_bar(stat = "identity", width = 0.6) +
+   coord_cartesian(ylim = c(0,1)) +
+   geom_text(aes(label = round(Accuracy,3)),
+             vjust = -0.5, size = 5) +
+   ggtitle("Impact of Data Preprocessing on Prediction Accuracy") +
+   ylab("Accuracy") +
+   xlab("Model") +
+   theme_minimal() +
+   theme(legend.position = "none")
> prob_raw <- attr(pred_raw, "probabilities")[,2]
> roc_raw <- roc(response = test$target,
+                predictor = prob_raw,
+                levels = c("0","1"),
+                direction = "<")
> prob_processed <- attr(pred_processed, "probabilities")[,2]
> roc_processed <- roc(response = test_scaled$target,
+                      predictor = prob_processed,
+                      levels = c("0","1"),
+                      direction = "<")
> plot(roc_raw, col="red", main="ROC Curve Comparison")
> plot(roc_processed, col="blue", add=TRUE)
> legend("bottomright",
+        legend=c("Raw Data","Preprocessed Data"),
+        col=c("red","blue"),
+        lwd=2)
> cat("\nAUC Raw:", round(auc(roc_raw),3), "\n")

AUC Raw: 0.102
> cat("AUC Processed:", round(auc(roc_processed),3), "\n")
AUC Processed: 0.898
> cat("\nConfusion Matrix - Raw Data\n")

Confusion Matrix - Raw Data
> print(cm_raw)
```

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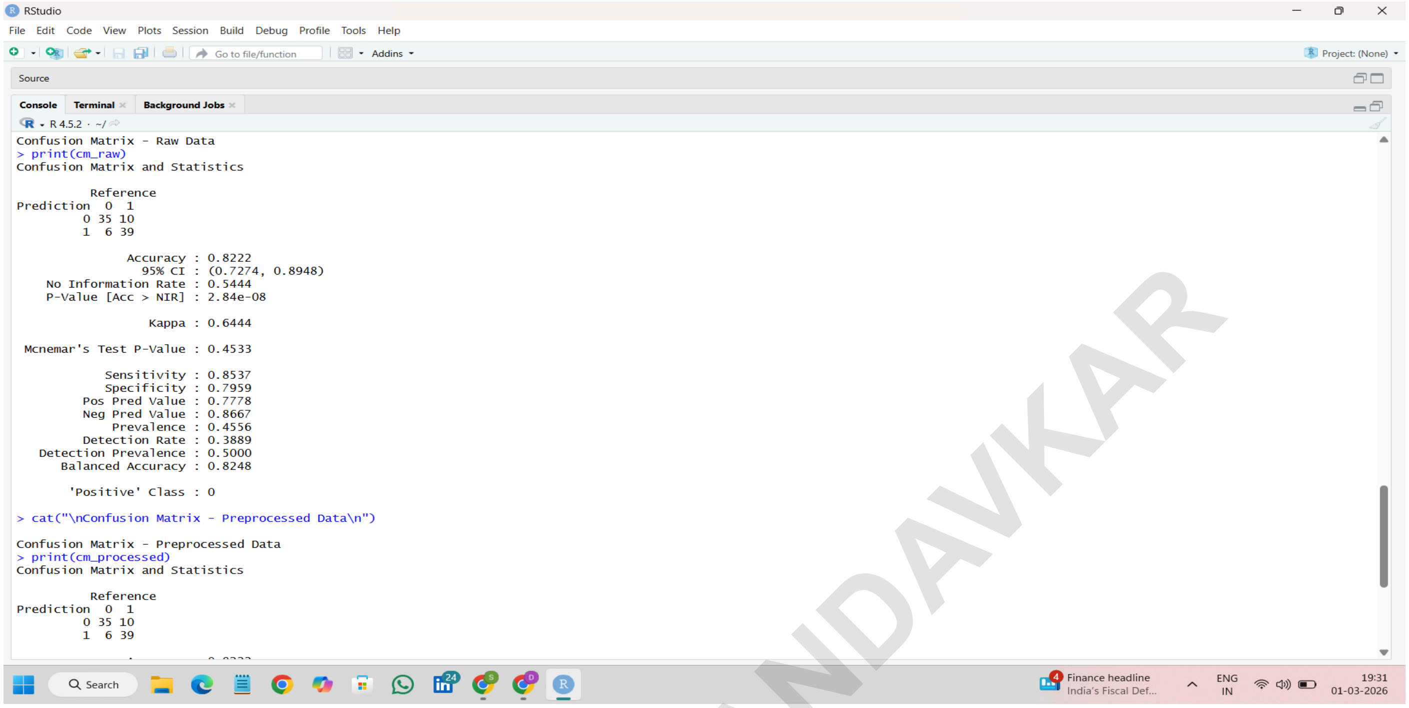
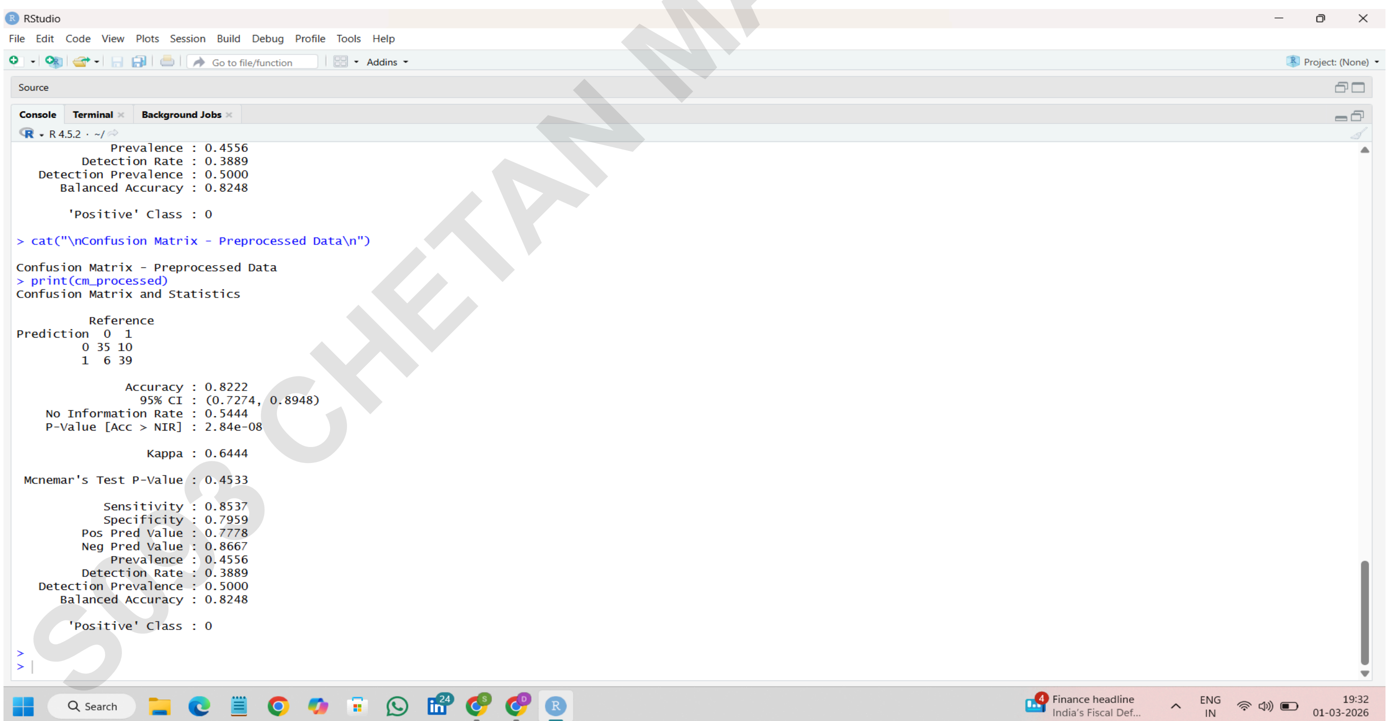


```
Confusion Matrix - Raw Data
> print(cm_raw)
Confusion Matrix and Statistics

          Reference
Prediction  0  1
         0 35 10
         1  6 39

               Accuracy : 0.8222
                 95% CI : (0.7274, 0.8948)
    No Information Rate : 0.5444
    P-Value [Acc > NIR] : 2.84e-08

                  Kappa : 0.6444

 Mcnemar's Test P-Value : 0.4533

            Sensitivity : 0.8537
            Specificity : 0.7959
         Pos Pred Value : 0.7778
         Neg Pred Value : 0.8667
             Prevalence : 0.4556
         Detection Rate : 0.3889
   Detection Prevalence : 0.5000
      Balanced Accuracy : 0.8248

       'Positive' Class : 0

> cat("\nConfusion Matrix - Preprocessed Data\n")

Confusion Matrix - Preprocessed Data
> print(cm_processed)
Confusion Matrix and Statistics

          Reference
Prediction  0  1
         0 35 10
         1  6 39
```



```
             Prevalence : 0.4556
         Detection Rate : 0.3889
   Detection Prevalence : 0.5000
      Balanced Accuracy : 0.8248

       'Positive' Class : 0

> cat("\nConfusion Matrix - Preprocessed Data\n")

Confusion Matrix - Preprocessed Data
> print(cm_processed)
Confusion Matrix and Statistics

          Reference
Prediction  0  1
         0 35 10
         1  6 39

               Accuracy : 0.8222
                 95% CI : (0.7274, 0.8948)
    No Information Rate : 0.5444
    P-Value [Acc > NIR] : 2.84e-08

                  Kappa : 0.6444

 Mcnemar's Test P-Value : 0.4533

            Sensitivity : 0.8537
            Specificity : 0.7959
         Pos Pred Value : 0.7778
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             Prevalence : 0.4556
         Detection Rate : 0.3889
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       'Positive' Class : 0

>
>
```


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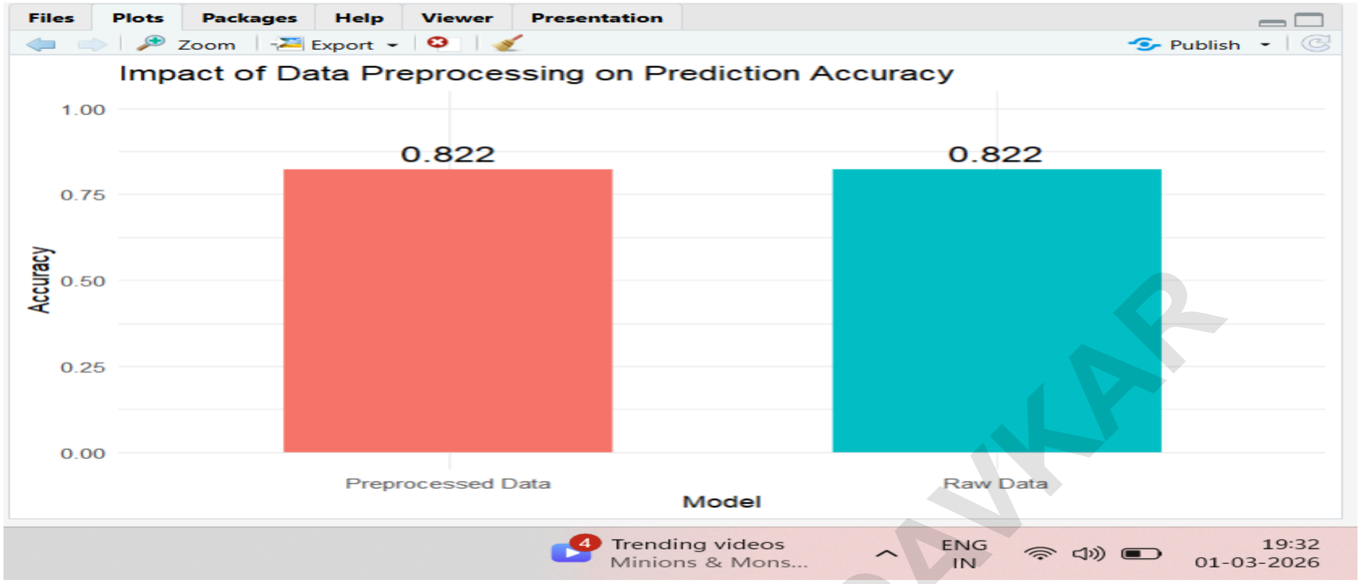

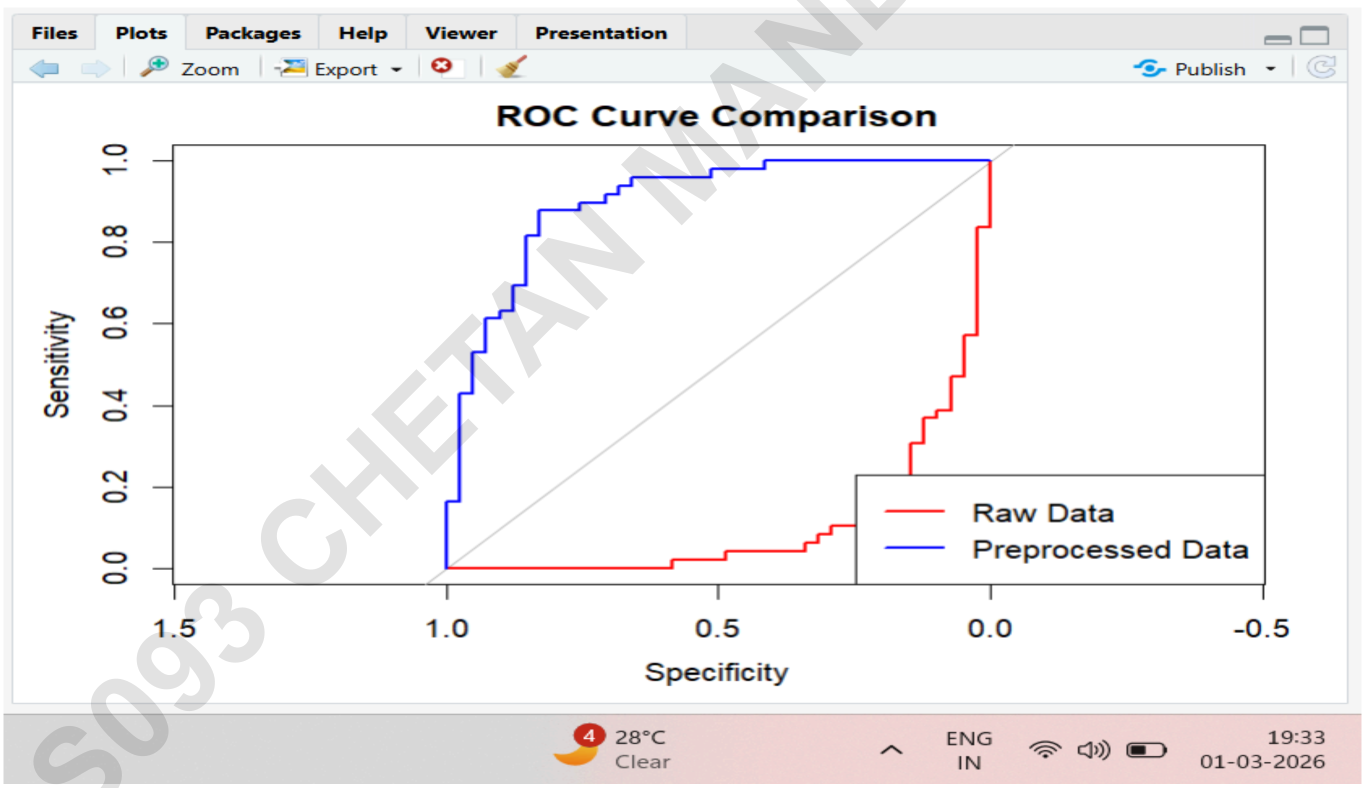

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